

Vibe-IC — ALL Steps: Phase → Stage → Step (v2.2.0)

One ordered list of every flow step, organized as **Phase → Stage → Step**, with a single continuous number **1 → 33 across Stages 1–4** (starting at Stage 1, Spec-to-RTL). Phase 1’ s doc-generation steps are lettered **D1–D5** (pre-flow, not in the 1→33 count). Two parallel tracks — Analog (A1–A9) and Mixed-signal (M1–M4) — run alongside. Source of truth = the runners; finer code-level markers are auto-generated in `FLOW_STEPS_GENERATED.md`.

Phase → Stage map

- **Phase 1** — Spec & Documents → two convergent input paths: **Agent path** (PM Agent • IC Expert Agent) • **doc-gen path** (D1–D5)
- **Phase 2** — RTL → Synthesis → Stage 1 (RTL+Verify) • Stage 2 (Synthesis+DFT)
- **Phase 3** — Physical → Tapeout → Stage 3 (Physical+Sign-off) • Stage 4 (Output+Tapeout)
- **Parallel** — Analog A1–A9 • Mixed-signal M1–M4

Phase 1 — Spec & Documents

Phase 1 has **two convergent input paths** that both yield the same L1–L23 layer documents. There is intentionally **no Phase 1a/1b sub-numbering** — per RFC v2.0 § 8.2 the entry difference is just an input subdirectory, not a phase split, since both paths produce L1–L13:

- `phase1/input_prompt/` — free text / natural language → the **Agent path** (PM Agent + IC Expert Agent).
- `phase1/input_doc/` — vendor docs / structured YAML → the **deterministic doc-gen path** (D1–D5).

Both converge on the L1–L23 docs that feed Phase 2.

Agent path — `phase1/input_prompt/` • 2 Agent Skills

When the input is free text, two agents drive Phase 1. The PM Agent faces the user; the IC Expert Agent works behind it and never talks to the user directly.

#	Agent Skill	Role	Faces user?
1	PM Agent (<code>agents/pm-agent.md</code>)	Natural-language front door: NL-ingest → gap dialogue (one question at a time, in the user’ s own words) → confirm the fact graph → hand off. Turns user product-talk into L1–L9 facts.	✓ yes
2	IC Expert Agent (<code>agents/ic-expert-agent.md</code>)	Silicon reviewer behind the PM Agent: reviews every layer for technical completeness, fills auto-decided defaults (with <code>auto_decided</code> + <code>reasoning trace</code>), cross-	✗ no — only via PM Agent

#	Agent Skill	Role	Faces user?
		checks layers (L5↔L4 pins, L6↔L5 signals, L9↔L5+L6+L8), applies design conservatism. This is the plugin's IC-expertise store — its per-layer review checklists are where domain knowledge accumulates (every benchmark-enhancement-capture Bucket-B recovery lands here).	

Hand-off chain: user free-text → **PM Agent** (NL-ingest + gap dialogue) → fact graph → **IC Expert Agent** (review / fill / cross-layer consistency) → finalized L-docs. The IC Expert Agent is where the plugin compounds silicon knowledge over time; the PM Agent keeps the non-expert user in plain language throughout. The handed-off facts then flow through the same doc-gen extractors below.

Deterministic doc-gen path — phase1/input_doc/ • D1–D5 (pre-flow • not in the 1→33 count)

#	Step	Tool / How
D1	Ingest & text extraction (prompt or vendor docs → input_doc/)	phase1_doc_one_shot_runner
D2	Generate L1–L13 core design-layer docs	deterministic extractors
D3	Generate L14–L23 protocol / timing / skeleton docs	overlay extractors
D4	Protocol-class synthesis dispatch (81 classes)	is_<proto> + <proto>_synth
D5	Coverage / parity report	phase1 parity report

Phase 2 — RTL → Synthesis

Stage 1 — RTL Generation & Verification

#	Step	Tool / How
1	Spec-to-RTL (author RTL from L-docs)	spec-to-rtl skill
2	Lint	eda_lint + hygiene gates
3	CDC / RDC check	cdc-check
4	Simulation	testbench-gen + eda_simulate
5	Formal verification	formal-verify + assertion-gen
6	FPGA early prototype	

#	Step	Tool / How
		eda_fpga_compile / eda_fpga_program → .sof

Stage 2 — Synthesis & DFT

#	Step	Tool / How
7	Constraint setup	constraint-gen → *.sdc
8	SDC validation	SDC lint
9	Synthesis (Yosys)	eda_synth + synth-doctor
10	Pre-layout STA	eda_sta_mcorner (SS/TT/FF)
11	DFT insertion (scan + ATPG)	dft-insert + atpg + eda_dft
12	Post-DFT optimization	resynth / buffering
13	Equivalence check (LEC)	equivalence-check + Yosys equiv

Phase 3 — Physical Design → Tapeout

Stage 3 — Physical Design & Sign-off

#	Step	Tool / How
14	Floorplan + PDN	eda_pnr (init)
15	Clock planning	clock-planning skill
16	Placement (global + detailed)	eda_pnr
17	CTS	eda_pnr enable_cts=true
18	Post-CTS hold fixing	repair_timing -hold
19	Routing (global + detailed)	eda_pnr enable_detailed_route=true
20	Parasitic extraction → SPEF	OpenRCX extract_parasitics
21	Post-route STA (MMMC)	eda_sta_mcorner
22	IR drop	OpenROAD PSM analyze_power_grid
23	EM check	PSM -enable_em
24	Antenna check	OpenROAD check_antennas
25	Signal integrity (crosstalk)	SPEF coupling-cap screen
26	Post-layout gate-level sim (+SDF)	eda_simulate
27	Physical verification (DRC / LVS / ERC + PERC-equivalent)	KLayout DRC + LVS sign-off chain + Magic ERC + perc_equivalent
28	ECO repair loop	eco-plan

Stage 4 — Output & Tapeout

#	Step	Tool / How
29	Power analysis	pre + post layout

#	Step	Tool / How
30	Metal fill (density fill)	OpenROAD filler_placement
31	Tapeout checklist	tapeout-checklist + signoff_audit
32	GDSII output	eda_gds + def2gds
33	FPGA final sign-off	recompile + on-board test + attestation

Parallel tracks

Analog A1–A9 (analog_one_shot_runner.py , runs alongside Stages 1–3)

#	Step	Tool / How
A1	spec_extract	→ A1_spec.json
A2	topology_select	analog-topology-select → A2_topology.json
A3	netlist_gen	→ <block>.sp
A4	corner_sweep	ams-sim → A4_corners.json
A5	layout (Magic)	→ A5_layout.json
A6	per-block physical verification (DRC + LVS)	analog_a6_block_pv_check
A7	post-layout resim	→ A7_postsim.json
A8	hardmacro_gen	→ {.lef,.lib,.gds,.v} (feeds Stage 3)
A9	hw_verify (HIL) / co-sim	→ A9_hw_verify.json

Mixed-signal M1–M4 (mixed-signal-cosim skill, no dedicated runner)

#	Step	Tool / How
M1	top merge	mixed_signal_m1_top_merge_check.py
M2	co-sim setup	mixed-signal-cosim skill
M3	co-sim run	mixed-signal-cosim skill
M4	integration verify	mixed-signal-cosim skill

Totals

Phase	Stages	Steps
Phase 1 — Spec & Documents	two input paths (Agent • doc-gen)	D1–D5 + PM Agent • IC Expert Agent
Phase 2 — RTL → Synthesis	Stage 1 • Stage 2	1–13
Phase 3 — Physical → Tapeout	Stage 3 • Stage 4	14–33
Parallel	Analog • Mixed-signal	A1–A9 • M1–M4

33 sequential steps (Stage 1: 1–6 • Stage 2: 7–13 • Stage 3: 14–28 • Stage 4: 29–33), plus **Phase 1** (two input paths: the Agent path — PM Agent • IC Expert Agent — and the doc-gen path D1–D5) and the two parallel tracks.

Pre-flight: P0 (mcp_server_health_check, eda_doctor). Orchestrator vibe_ic_one_shot_runner.py runs Phase 1 → Phase 2 → Analog → Phase 3.

Detail beyond this summary: the runner's live code-level markers in FLOW_STEPS_GENERATED.md; the LVS sign-off chain + PERC-equivalent coverage in PERC_SIGNOFF_MEMO.md. 繁中版見 ALL_STEPS_v2.2.0.zh-TW.md.